Exercise 7

<u>Modification of exercise 6</u>

1) Modify the previous exercise assuming that the subject is not breathing (therefore, he is under sedation) and that the muscle contribution is zero. Imagine artificial ventilation by providing a pressure at the mouth P_{vent} (t) that allows for alveolar ventilation values, alveolar volumes, and pressures similar to natural ones.

2) Assume the patient has residual free breathing (for example, P_{mus} varies between 0 and - 2 cmH_2O). Add ventilation to the mouth to aid total ventilation. What is occurring?

3) Ventilation regulation operates by varying the amplitude and frequency of the breath. Suppose you change the amplitude of the P_{mus} (for example, by increasing it up to + 100%) or increase the respiratory rate (up to + 100%) and evaluate the dependence of the parameter on the parameter change.

4) Use the following more precise form of muscle pressure during free breathing

$$P_{mus}(t) = \begin{cases} \dfrac{-P_{mus,min}}{T_I \cdot T_E} \cdot t^2 + \dfrac{P_{mus,min} \cdot T}{T_I \cdot T_E} \cdot t & \text{se} \quad t \in [0, T_I] \\ \dfrac{P_{mus,min}}{1 - e^{-\frac{T_E}{\tau}}} \cdot \left(e^{-\frac{(t-T_I)}{\tau}} - e^{-\frac{T_E}{\tau}} \right) & \text{se} \quad t \in [T_I, T] \end{cases}$$

where T_I, T_E, and T denote the inspiratory, expiratory, and total period, respectively. Use a ratio $T_I/T = 0.4$ and □ = $T_E/5$. (advice, build the waveform for one period, and then repeat it identically to make a periodic waveform)

5) [optional – quite difficult]

Design a controller that makes $P_{vent}(t)$ vary through the following differential equation.

$$\tau_{control} \frac{dP_{Vent}(t)}{dt} = -P_{Vent}(t) + k_{control}\left(\dot{V}_{ref}(t) - \dot{V}(t)\right)$$

where $\dot{V}_{ref}(t)$ indicates the desired value for ventilation (function of time). Identify the values of □$_{control}$ and $k_{control}$ that make the ventilation relatively stable (range 6-8.5 l / min) despite a change in the patient's respiratory activity (that is, despite a change in the parameter $P_{mus,min}$).

This exercise extends the respiratory mechanics model of Exercise 6 by replacing or combining the natural respiratory muscle action with an external mouth pressure generated by a ventilator. The state equations remain the same as in the previous model; what changes is the way the system is driven. In spontaneous breathing the driving term is the muscle pressure Pmus(t), whereas during artificial ventilation the forcing term is the pressure applied at the mouth, here indicated as Pvent(t) or Pm(t). The goal is to compare spontaneous breathing, fully controlled mechanical ventilation, mixed patient-plus-ventilator breathing, and a simple feedback controller for ventilator assistance.

1. Artificial ventilation with zero muscle activity

When the patient is sedated, spontaneous breathing is absent and therefore Pmus(t) = 0. In this case the pleural pressure becomes Ppl = P5, and ventilation must be produced entirely by the mouth pressure supplied by the ventilator. The model equations are the same as in Exercise 6, except that the mouth pressure is no longer zero. The main auxiliary pressures are Pl = P1, Ppl = P5 + Pmus, Pt = P2 + Ppl, Pb = P3 + Ppl, and PA = P4 + Ppl. The airflow entering the system is Vdot = (Pvent - Pl)/Rml and the alveolar flow is VdotA = (Pb - PA)/RbA.

To reproduce a breathing pattern similar to natural breathing, the code applies a periodic mouth pressure of the form Pvent(t) = Avent·cos(2πt/T) - Avent, with period T = 5 s. By choosing a suitable amplitude, approximately 2–3 cmH2O in the provided simulations, the resulting alveolar pressure, alveolar volume, and minute ventilation become comparable to those observed during free breathing. Therefore, controlled mechanical ventilation can replace the muscular pump and maintain physiologically reasonable tidal excursions even when the patient does not breathe spontaneously.

2. Assisted ventilation with residual spontaneous breathing

If the patient still has residual respiratory activity, for example with Pmus varying between 0 and about -2 cmH2O, the ventilator does not fully replace the respiratory muscles but assists them. In this situation the total motion of the system depends strongly on the relative phase between Pmus and Pvent. If the ventilator support is synchronized with the patient effort, the two actions sum constructively: inspiratory flow increases, alveolar volume rises more, and minute ventilation improves. If instead the ventilator acts out of phase with the patient, patient–ventilator asynchrony appears. In that case the mouth pressure can partially oppose the inspiratory effort, producing irregular pressures and less efficient ventilation.

This is exactly the physiological meaning of assisted ventilation: the ventilator should amplify the patient effort, not fight against it. The mixed-ventilation simulation in the MATLAB script illustrates this point well, because the combination of a mouth pressure and a shifted muscle pressure can either improve or worsen the resulting flow and volume depending on the phase delay.

3. Effect of changing amplitude and respiratory frequency

Ventilation regulation can be represented by changing the amplitude and the frequency of Pmus. Increasing the amplitude of Pmus means stronger inspiratory muscle activity. In general, a larger amplitude produces a larger change in pleural pressure, which in turn generates a larger tidal

volume and larger alveolar ventilation. Therefore, tidal volume increases approximately with inspiratory effort, at least within the physiological range of the model.

Increasing the respiratory frequency has a different effect. Minute ventilation is equal to tidal volume multiplied by breathing frequency, so if frequency increases moderately, minute ventilation increases. However, if the frequency becomes too high, the system has less time to fill and empty during each cycle, and the tidal volume may decrease. For this reason the dependence on frequency is not purely linear: at first ventilation increases, but beyond a certain point efficiency decreases because each breath becomes shallower.

4. More realistic muscle-pressure waveform

Instead of a sinusoidal muscle pressure, the exercise proposes a more realistic respiratory waveform. In the code this is generated over one period and then repeated periodically. If tt is the local time inside one breathing cycle, the inspiratory phase is described by a parabolic law for $0 \leq tt < TI$, while the expiratory phase is described by an exponential decay for $TI \leq tt \leq T$. With $TI/T = 0.4$, $TE = T - TI$, and $\tau = TE/5$, the waveform becomes:

For inspiration: $Pmus1(tt) = -Amus/(TI \cdot TE) \cdot tt^2 + Amus \cdot T/(TI \cdot TE) \cdot tt$

For expiration: $Pmus2(tt) = Amus/(1 - \exp(-TE/\tau)) \cdot (\exp(-(tt - TI)/\tau) - \exp(-TE/\tau))$

The complete waveform is $Pmus(tt) = Pmus1(tt) + Pmus2(tt)$. Compared with the sinusoid, this profile is more physiological because inspiration is active and relatively shorter, whereas expiration is passive and slower. As a result, the pressure, flow, and volume curves look more realistic.

5. Optional feedback controller for mouth pressure

The optional part introduces a controller that automatically adjusts the mouth pressure in order to keep ventilation close to a desired value. In the MATLAB implementation the controller is written as a first-order differential equation:

$dPvent/dt = [\ -Pvent + kcontrol \cdot (Vdotref - Vdot)\ ] / \tau control$

where Vdotref is the desired flow profile, Vdot is the actual mouth flow, τcontrol is the controller time constant, and kcontrol is the proportional gain. This law increases support when the measured ventilation is below the target and reduces support when ventilation is excessive.

From the tested values in the code, a reasonable compromise is obtained with τcontrol around 2 s and kcontrol around 20–25. With these values the ventilator assistance is responsive but not excessively aggressive, and the resulting ventilation remains in a relatively stable range of about 6–8.5 L/min even when the spontaneous respiratory effort changes. If kcontrol is too small, the ventilator reacts weakly and ventilation may drop too much. If kcontrol is too large, the system may become oscillatory and uncomfortable because the ventilator overreacts to small flow errors.

Conclusion

In conclusion, Exercise 7 shows that the same respiratory mechanics model can represent natural breathing, controlled mechanical ventilation, assisted ventilation, and simple closed-loop ventilatory support. Artificial ventilation can replace the muscular pump when Pmus is zero,

assisted ventilation can improve breathing only if it is synchronized with the patient effort, and both tidal volume and minute ventilation depend on the amplitude and frequency of the respiratory drive. A more realistic muscle-pressure waveform improves the physiological interpretation of the results, while a feedback controller acting on mouth pressure can keep ventilation near a desired range despite variations in patient effort.